

Final subcircuits

Definition:

- A final subcircuit is an outgoing circuit connected to a distribution board and intended to supply electrical energy to a current using ~~application~~ apparatus, either directly or through socket-outlets or fused spur boxes.

IEE Regs requirement:

- (i) - Where an installation comprises more than one final subcircuit, each circuit shall be connected to a separate way in a distribution board.
- (ii) The wiring of each final subcircuit shall be electrically separate from that of every other final subcircuit.
- (iii) To facilitate disconnection of each final subcircuit for testing, the neutral conductors shall be connected at the distribution board in the same order as that in which the Live conductors are connected to the fuses or circuit-breakers.

~~(iii) All to~~

Construction of a final subcircuit.

— Can vary from a pair of 1.0mm^2 cables feeding one lamp, to a heavy 3-core PILC cable feeding a large motor from a circuit breaker located at a factory switchboard.

Electricity Supply Regulations requirements - ~~regulations~~

"— all conductors and apparatus must be of sufficient size and power for the work they are called on to do, and so constructed, installed and protected as to prevent danger."

CONSUMER'S INTAKE POINT.

IEE Regs recommends that every Consumers installation shall be adequately controlled by switch gear which are readily accessible to the Consumer and which incorporates the following 3 essential features:

- i) Means of Isolation
- ii) Means of excess-Current protection.
- iii) Means of earth-leakage protection.

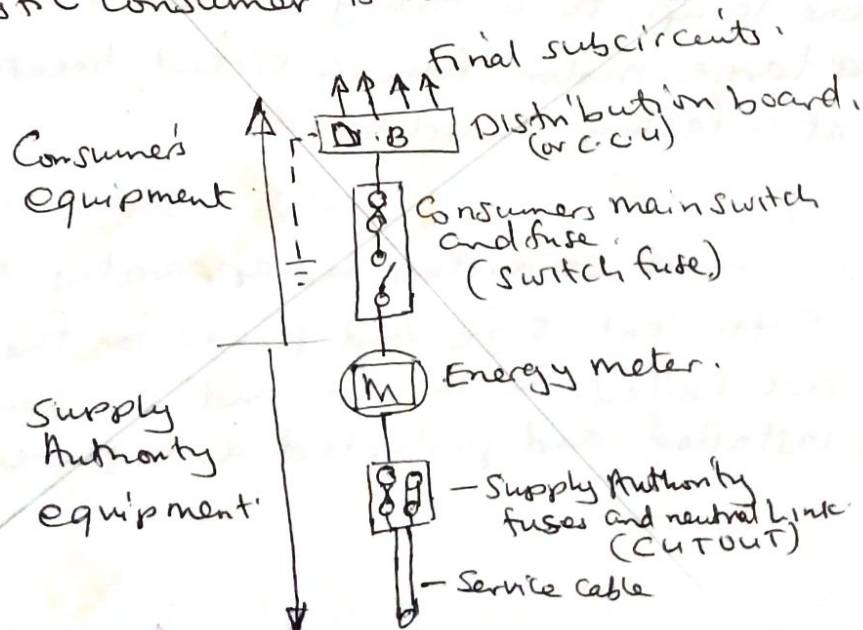
Note: The size and type of main switchgear to be installed depends on the ~~premise~~ type of premises and anticipated load.

For average domestic premises the service cable is 2-core P.I.C or PVC insulated. —

The cable is suitable for 20kW load at 240-v single-phase and is 16mm² C.S.A.

Industrial supplies unless they are very small loads, are provided with three-phase four wire services usually at 415 v b/n phases,

A typical supply-intake position for a domestic consumer is as illustrated below.



Types / Groups OF FINAL SUBJECTS.

There are five general groups of final subcircuits:

- (1) Rated at not more than 15A.
- 2) Rated Over 15A.
- 3) Rated over 15A but Confined to feeding 13A socket-outlets with fused plugs.
- (4) Circuits feeding fluorescent and other discharge lamps
- (5) Circuits feeding motors.

An industrial installation may comprise all the five ~~groups~~ types but a domestic premises may have only 1, 2 and 3.

INSTALLATION PLANNING.

@ Domestic installation

Points to consider includes:

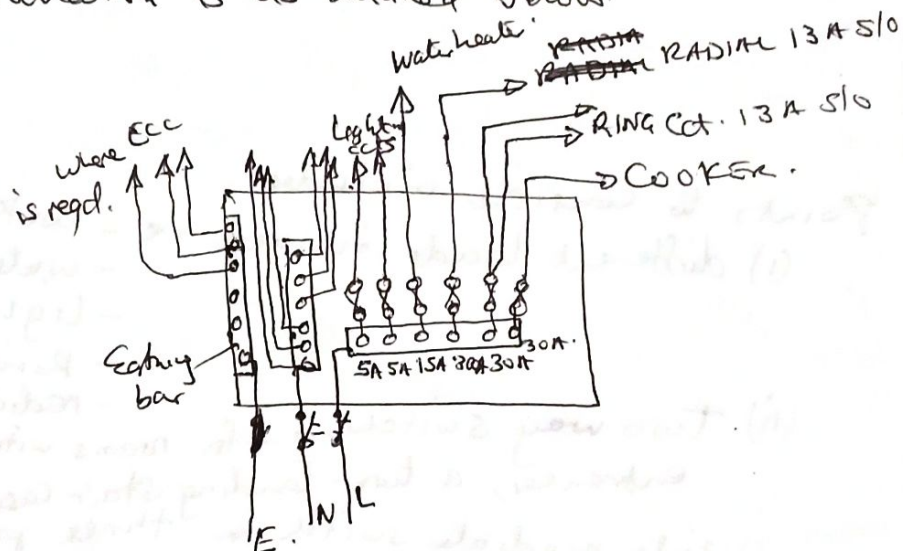
- (i) different loads involved i.e.
 - Cooker
 - water heater
 - lights
 - Ring ckt.
 - radial ckt
- (ii) Two way switching - for rooms with two entrances, a two-landing stair case
- (iii) Intermediate switch for three position control i.e. climbing stairs, long corridors
- (iv) need for two or more lighting circuits overlapping.

note: In the interest of good planning it is undesirable that the whole of the fixed lighting of an installation should be supplied from one final subcircuit. Reason: If one lighting ckt. fails the house is not plunged into darkness. Thus a slight overlap of lighting circuits should be considered - i.e. to wire one lighting point

to from one circuit within the wiring area of the other circuit.

Essentially lighting in houses should be regarded as an important aspect of interior decoration, as well as supplying lighting on a purely functional basis. That is why living rooms and ~~bedrooms~~ bedrooms may require wall mounted as well as ceiling mounted lamps. Sometimes local lighting is ~~required~~ recommended at the working surfaces or tables at kitchen or living rooms. In planning localised lighting is afforded by allowing provision of socket outlets at the proposed positions.

Typical Consumer unit arrangement for domestic installation is as outlined below:



8) Commercial Installation

The difference b/n domestic and commercial installations is that Commercial buildings are built as basic shells with the final requirements for lighting and other circuits not known until the office tenants sign their lease. The lighting in such buildings is general, special and depends on building services.

FINAL SUBCIRCUITS

Definition — is an outgoing circuit connected to a distribution board and intended to supply electrical energy to current using appliance (apparatus) either directly or through socket outlets or fused spur boxes.

TYPES OF FINAL SUBCIRCUITS

There are five general groups of final subcircuits

- 1) Rated at not more than 15A.
- 2) Rated over 15A.
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- 5) Circuit feeding motors.

Note:

An industrial installation may have all five types, a domestic installation may have only 1, 2, and 3.

IEE Regulation Requirements for each type of final subcircuit

(1) Circuits rated under 15A.

— may ~~serve~~ feed an unlimited number of points provided that the total 'current demand' does not exceed 15A.

— Various types of points in this category includes 15A, 13A, 5A and 2A socket outlets, lighting ~~outlet~~ outlets, stationary appliances and certain loads which may be

neglected because their current demand is negligible (eg clocks, bell transformers, electric-shower supply units) provided that their rating is not greater than 15A.

— The current rating of the cable must not be exceeded.

Note: For domestic installations lighting fuse rating is 5A. Industrial lighting ccts are usually rated at 15A, because of the higher wattage of the lamp used.

(II) Circuits Rated over 15A:

— With two exemptions, circuits rated at over 15A should not serve more than one point. The exemptions are circuits which feed 13A socket-outlets and cooker circuits.

(III) Circuits rated for 13A socket-outlets:

— Two types of circuits are recognised;
(i) the radial and
(ii) the ring.

Regs A31 to A40

- Each socket-outlet of a two-gang or multiple socket-outlet is to be counted as one.
- Stationary appliances, permanently connected to a radial or ring cct, must be protected by a fuse not exceeding 13A rating and controlled by a switch or a circuit breaker.
- Except where a ring cct is run throughout in metallic conduit, ducts or trunking, the ecc shall be run in the form of a ring, having both ends connected to earth at the distribution board (or its equivalent).
- The total number of spurs shall not exceed the total number of socket outlets and stationary appliances connected directly to the ring.

13A S/O Continue.

- Fused Spurs from ring circuits must be connected through fused spur boxes, and the rating of the fuse must ~~be~~ not exceed the current rating of the cable forming the spur, and in any event must not exceed 13A.

Circuits Feeding discharge Lamps

- if a switch is not specifically designed to ~~break~~ ^{break} ~~control~~ highly inductive ^{loads} ~~circuits~~ (quick-make, slow break), it should have a current rating of not less than twice the total steady current which it is required to carry.

Circuits feeding motors:

- Cable which carry the starting, accelerating and load currents of a motor must be rated at least to the full-load current rating of the motor.
- More than one motor may be connected to a 15A final subcircuit, provided that the aggregate full-load rating of the motors does not exceed 15A.
- If a motor takes more than 15A full-load current it should be fed from its own final subcircuit.

FINAL SUBCCT PROTECTION

- by means of fuses, circuit breakers or miniature circuit breakers located at switchboards and distribution boards.
- The protection is for over-currents caused by

Short-circuits between conductors, between cabs and Earth or overloads.

Overview

Reasons for different final subcircuits in a building:

- In case of supply failure the room is not plugged into total darkness (for lighting circuits).
(ie to allow an alternative lighting point(s) in case one
(ie to provide alternative sources of power in case one fails ie for lighting and power circuits).
- to avoid circuit overloading and create load balancing.

- Provision of adequate socket outlets is a particular problem and should be flexible enough to cater for possible load increase i.e. increase of additional Computers, typewriters etc.

c) Industrial Premises

- require ~~installations~~ lighting installations which cater for the intensity required for the process to be carried out.
- The lighting installation within the factory area should be wired with a system which will allow extensive alteration and significant addition with ease of maintenance and re-wiring when necessary.

d) Special type of premises

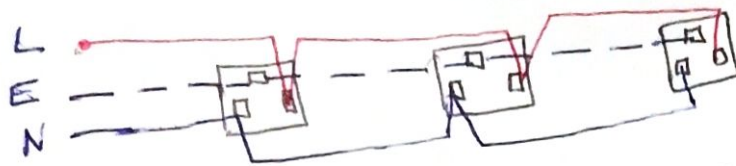
- includes hospitals, theatres, cinemas, hotels, schools and buildings of historic interest
- Like hospital, specific code of practice should be followed as laid down by the Ministry of Health.
- In places of entertainment, secondary lighting installations are required. (Emergency lighting system).

Assignment

Briefly ~~summarise~~ discuss the IEE regulations requirements for each of the five types of final subcircuits.

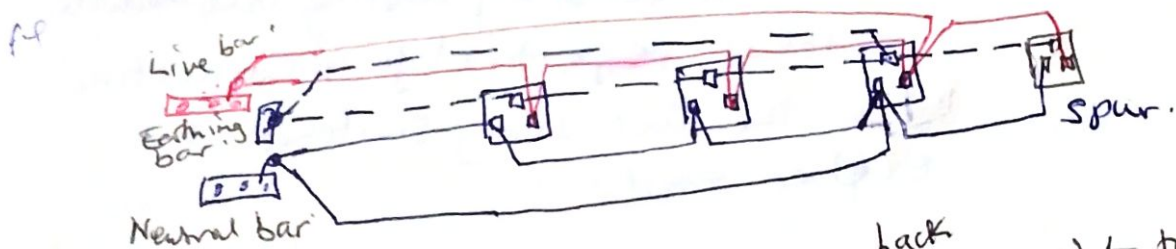
89. Typical Final Subcircuits. in common use.

(1) Radial 13A. Socket outlet.



— power supply is linear ~~for~~ looped ^{from one} socket-outlet to another.

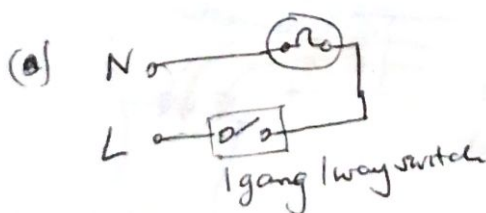
(2) 13A Ring socket outlet with a spur.



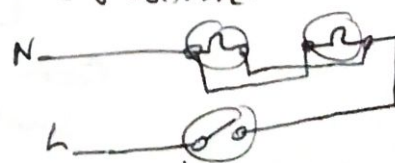
— The Conductors are looped ^{back} (returned) to the same way at the distribution board and socket outlets so that the supply is at both ends of the circuit.

(3) LIGHTING CIRCUITS.

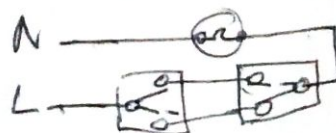
(a) one Lamp controlled from one position.



(b) Two lamps controlled from one position.

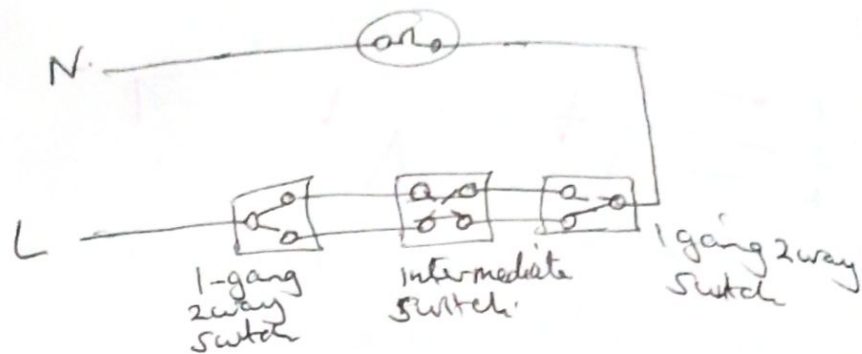


(c) A lamp controlled from two independent positions.



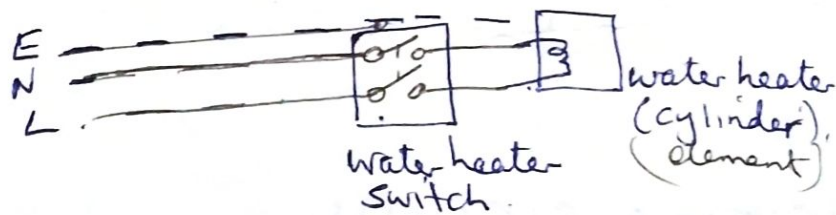
requires two, 2 way switches.

(d) A lamp controlled from three independent positions.



(e) A lamp controlled from more than three positions; -- achieved by adding intermediate switches respectively in conjunction with the two 2-way switches at either end.

(A) Water heater circuit.



(5) Cooker circuit

